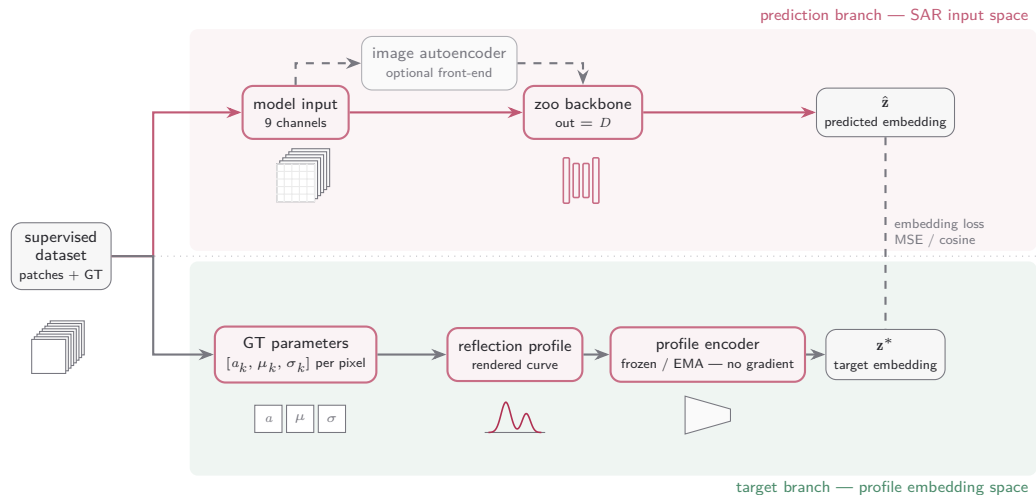


DLR-TomoSAR — JEPA pretraining direction

predict in representation space · results pending

July 2026

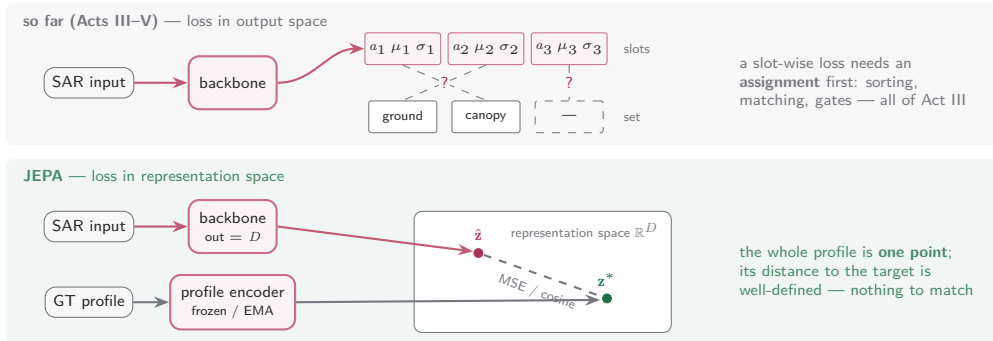
What JEPA does



A network learns to predict, from the reduced stack alone, the **embedding** that a separately-trained profile autoencoder assigns to the ground-truth reflection profile.

JEPA: predict in representation space

The supervision is unchanged — what changes is *where* the error is measured: not between raw outputs, but between learned **embeddings** of whole profiles (I-JEPA, Assran et al., CVPR 2023).



- **Act III dissolves by construction:** an embedding of the whole profile has no slots — nothing to sort, match, or gate.
- **A friendlier target:** low-dimensional, LayerNorm-normalised, and nearby embeddings decode to similar curves — the loss compares *profiles*, not coordinate arrays.
- **A decoupled target:** no gradient reaches the frozen / EMA profile encoder, so the predictor cannot bend the representation toward a trivial loss.

Instantiation for TomoSAR

Separates *what a profile is* from *predicting it from SAR*:

- **Profile autoencoder**: per-pixel, position-agnostic encoder-decoder over the profile; seven variants (MLP, Conv1D, Transformer1D, ResMLP, TCN, GRU, Conv-Attention); embedding $D = 24$, LayerNorm-normalised.
- **JEPA predictor**: any zoo backbone rebuilt with out = D ; target $\mathbf{z}^*$ via stop-gradient, live, or EMA encoder (decay 0.996 — the I-JEPA scheme).
- **Optional image-autoencoder front-end** on the SAR stack — three coupling combinations under one pipeline.
- **Loss**: embedding match (MSE/cosine) + curve-reconstruction anchor through the frozen decoder, so $\hat{\mathbf{z}}$ stays decodable to a physical profile.
- **Inference**: embedding decoded to a curve, scored on the same curve metrics as the supervised models — directly comparable to Act V.

Why it matters — and status

- **Decoupling**: the profile embedding is trained cheaply and reused across backbones.
- **Second path around the label ceiling**: physics losses constrain the *output* space, JEPA restructures the *target* space — they compose.
- **Pretraining bridge**: an embedding-predictive objective is the natural route toward training on unlabeled stacks.

Status

Full pipeline implemented end-to-end (training, coupling modes, inference scoring); **systematic training studies pending**.